

Does Deterministic Generate–Validate–Repair Reduce Unsafe LLM-Generated Network Changes? A Confirmatory Paired Study

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Abstract—We preregistered and executed a sealed paired confirmatory study ($n = 200$ intent→configuration tasks) to test whether a deterministic generate–validate–repair (GVR) pipeline that consults a deterministic NetOps validator reduces validator-detected unsafe or semantically incorrect network changes relative to an LLM-only generation pipeline. The run used a single hosted model (gpt-5-mini) with adapter-enforced shared-initial generation semantics and a primary deterministic validator (deterministic_netops_validator_v5). Execution completed all planned pairs (completed_pairs = 200) consuming 200 model calls (one shared initial generation per task); no repair calls were issued because every initial candidate passed the validator. Paired contingency: $n_{11} = 200$, $n_{10} = 0$, $n_{01} = 0$, $n_{00} = 0$; estimate = 0.0; exact McNemar two-sided $p = 1.0$; 95% bootstrap percentile CI = [0.0, 0.0] (10,000 resamples, seed = 1729). Because the baseline validator-detected unsafe incidence was 0% on this task bank, the preregistered $\geq 50\%$ relative-reduction hypothesis could not be meaningfully evaluated. We report the sealed design, execution accounting (start: 2026-08-30T11:50:34.530059Z; end: 2026-08-30T12:12:07.541650Z), the Mininet 10% hold-out (20 tasks) that produced zero validator–emulator disagreements, and an artifact-grounded discussion of operational implications and threats to validity (preregistration id: cnsms2026-gvr-effectiveness-02).

I. INTRODUCTION

Automating intent→configuration translation with generative language models can increase NetOps productivity but risks producing incorrect or unsafe changes that cause outages or policy violations. Prior work therefore proposes grounding, deterministic validation, and repair loops (generate–validate–repair, GVR) to detect and correct unsafe candidates before application. The operational benefit of automated repair is conditional on three factors: (i) baseline prevalence of validator-detectable failures from the chosen generator/prompting policy; (ii) validator coverage and fidelity; and (iii) the available repair budget and repair-prompt effectiveness. We preregistered a narrowly scoped confirmatory question: Does an automated, deterministic GVR pipeline that consults a deterministic NetOps validator materially reduce validator-detected unsafe or semantically incorrect network changes relative to an LLM-only generation pipeline on a curated 200-task intent→configuration bank? The study was designed as a paired experiment that fixes the generator output per task and isolates the marginal effect of invoking a validator+repair on the same candidate.

II. RELATED WORK

Surveys and empirical studies across LLM applications document factual errors, hallucinations, prompt sensitivity, and non-determinism; mitigation approaches include retrieval grounding, verifier-assisted repair, and constrained agent architectures. NetOps prototypes show feasibility of intent→configuration generation but typically lack instrumented, production-like failure-rate evaluations. Architectural proposals recommend combining LLM generation with deterministic validators (e.g., Batfish-style checkers) and with guarded agent patterns; red-team audits demonstrate configuration-level brittleness and tool-description injection vectors that can alter safety outcomes. Our study positions itself as an externally auditable, preregistered measurement that isolates the marginal effect of deterministic repair for a fixed model and sampling policy, addressing gaps about when GVR actually reduces validator-detected failures in practice.

III. METHODOLOGY

Preregistration and inferential plan. The study preregistration (cnsms2026-gvr-effectiveness-02) fixed a paired design with $n = 200$ tasks, one shared deterministic initial generation per task, and a repair budget of at most one automated repair call per task. The primary estimand was the paired_success_rate_difference_guarded_minus_baseline; the primary test was an exact two-sided McNemar test ($\alpha = 0.05$). Confidence intervals were computed via bootstrap percentile (10,000 resamples, bootstrap_seed = 1729). Power planning assumed a baseline unsafe incidence $\approx 30\%$ and a targeted 50% relative reduction; these assumptions were recorded in the preregistration and used only to justify sample size. The analysis executor was paired_binary_analysis_v1.

Task bank, validators, and holdout. We used a curated, machine-checkable intent→configuration task bank with 200 tasks across four topology clusters (enterprise, campus, datacenter, edge; 50 tasks each). Each task encodes an initial and expected final state, workflow constraints (make-before-break, must_be_down_for_fields, group-active minima), and a required_sequence. The primary deterministic checker (deterministic_netops_validator_v5) enforces syntactic and intent/constraint checks and emits structured diagnostics (valid flag, violation codes, parsed_command_count, required_command_count). A Mininet-based data-plane emulator was preregistered as a secondary disagreement de-

tector applied to a randomized 10% holdout (20 tasks) to detect validator–emulator mismatches; execution produced an `analysis/contamination_summary.csv` artifact recording zero disagreement flags for that holdout.

Adapter semantics, shared-initial design, and causal isolation. The `hosted_netops_gvr_v1` adapter enforced shared_initial generation semantics: for each task the system produced a single deterministic initial candidate that was reused for both baseline (LLM-only) and guarded (GVR) conditions. This design choice intentionally removes sampling variance across conditions so the paired comparison isolates the marginal effect of invoking the validator and (if needed) issuing a single repair prompt on the identical generator output. Adapter accounting (deterministic_reconciliation artifacts) records `hosted_model_call_event_count = 200` and `stage_counts = {shared_initial_generation: 200}`, confirming single shared invocations per task. By construction, any benefit attributable to repeated or stochastic re-sampling is excluded from the primary estimand; this is a preplanned trade-off to obtain a controlled causal contrast between (A) applying the generator output unchecked and (B) validating—and possibly repairing—that same output.

Repair policy, budgeting, and handling. The preregistered repair policy allowed at most one automated repair prompt to the same LLM when the primary validator reported violations. Repair prompts (when issued) and any repaired candidate were re-validated by `deterministic_netops_validator_v5` and recorded in `validator_trace` fields (`repair_applied`, `repair_prompt_sha256`, `final_configuration`). The execution contract limited maximum model calls to 400 (200 initial + up to 200 repairs); the adapter enforced this cap. Missingness/exclusion rules deterministically excluded pairs where both conditions failed due to infrastructure execution failures; tasks flagged for validator–emulator disagreement were excluded from the primary analysis per the `contamination_plan`, with deterministic sensitivity analyses preregistered for alternative treatments.

Randomization, seeds, and reproducibility artifacts. Task ordering was randomized with a fixed generation seed as recorded in the `task_manifest` entries (example fields: `generation_seed`). Analysis used a paired, shared-initial seed schedule and the bootstrap CI procedure used `bootstrap_seed = 1729` with 10,000 resamples (`analysis/analysis_log.jsonl` captures these parameters). The Mininet 10% holdout executed as preregistered and is summarized in `analysis/contamination_summary.csv` (20 tasks; zero disagreement flags). We disclose a documented reproducibility gap: the explicit labeled randomized selection seed identifying which tasks comprised the 10% Mininet holdout is not present in the labeled artifacts, preventing exact deterministic reconstruction of the drawn holdout despite the contamination summary reporting zero disagreements.

Statistical implementation and exact tests. The primary confirmatory test was implemented as an exact McNemar two-sided test on the 2×2 paired table (n_{11} , n_{10} , n_{01} , n_{00}) produced by `paired_binary_analysis_v1`. Where

the discordant counts were zero or small, exact inference is necessary; bootstrapped paired-difference percentile CIs were computed to reflect the empirical paired distribution under resampling (10,000 resamples; seed recorded). Secondary/exploratory analyses (preplanned) included logistic regressions (`repair_success ~ topology_cluster + initial_validator_failure_type + repair_iterations`) and descriptive summaries of repair iteration counts and model-call costs; these remained exploratory and were executed only when applicable data were generated.

Contamination detection and conservative primary analysis rule. The preregistered contamination plan used the Mininet emulator as an independent end-to-end check for a randomized holdout. Any pair where Batfish-style validator verdict $\neq$ Mininet verdict would be flagged and excluded from the primary confirmatory analysis to conservatively avoid reliance on a possibly incomplete primary validator. The analysis artifact `analysis/contamination_summary.csv` shows zero flags for the 20-task holdout; `deterministic_reconciliation.json` confirms `pair_accounting_consistent = true`. Because no contamination flags occurred, the preplanned conservative exclusion branch was not activated.

Execution accounting and archived artifacts. The run executed autonomously under the frozen plan (start: 2026-08-30T11:50:34.530059Z; end: 2026-08-30T12:12:07.541650Z). Model calls used = 200 (one shared initial generation per task); repair calls used = 0. Core archived artifacts relevant to reproducing and auditing the methodology include `execution/raw_results.jsonl`, `execution/model_configuration.json`, `execution/task_manifest.jsonl`, `analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/contamination_summary.csv`, `analysis/analysis_log.jsonl`, and `deterministic_reconciliation.json`. These artifacts capture adapter accounting, validator traces (validation_before/after objects with `parsed_command_count`, `required_command_count`, `violation_codes`), and the bootstrap parameters. All handling decisions (exclusions, missingness treatment) were deterministic and logged in `analysis/analysis_log.jsonl`.

IV. RESULTS

Execution completed all planned episodes and pairs. Accounting: `completed_episode_count = 400` (shared initial generation reused across conditions), `complete_pair_count = 200`, `model_calls_used = 200` (one shared initial generation per task), `maximum_model_calls = 400`, `repair_calls_used = 0`. Adapter/provider audit artifacts report `hosted_model_call_event_count = 200`, `completed_hosted_model_call_event_count = 200`, `failed_hosted_model_call_event_count = 0`, `cache_miss_count = 200`, and `stage_counts = {shared_initial_generation: 200}`, confirming the preregistered single shared initial invocation semantics.

Validator outcomes and paired contingency. Every initial candidate passed the primary deterministic validator: base-

line_success_count = 200/200 and guarded_success_count = 200/200 (analysis/condition_summary.csv). The paired contingency table is $n_{11} = 200$, $n_{10} = 0$, $n_{01} = 0$, $n_{00} = 0$ (analysis/paired_contingency_table.csv). Because there are no discordant pairs, the observed paired difference estimate = 0.0. The prespecified exact McNemar two-sided test yields $p = 1.0$. Bootstrap percentile confidence intervals (10,000 resamples, bootstrap_seed = 1729 as recorded in analysis/analysis_log.jsonl) produce a 95% CI = [0.0, 0.0]. Under the preregistered estimand and test the preregistered $\geq 50\%$ relative-reduction hypothesis is not evaluable on this sample because baseline validator failures are absent.

Repair and secondary outcomes. No validator failures were observed; consequently, the preregistered repair branch was never invoked and secondary_results are empty in the analysis bundle. The planned secondary estimands (average repair iterations per repaired task, GVR-introduced failure rate, model-call cost per successful repair) therefore have no empirical data in this run and are recorded as empty artifacts. The analysis/contamination_summary.csv (the preregistered Mininet holdout check) contains no flagged disagreements: the randomized 10% Mininet holdout (20 tasks) produced zero validator-emulator mismatches per that artifact.

Representative diagnostics and task difficulty. Representative per-task validator traces (e.g., execution/scoring artifacts for pair-000001...pair-000006) confirm that the shared initial generator outputs matched the task required_sequence and that validator_trace.validation_after.valid = true with violation_count = 0. These representative artifacts also show that the task bank exercised nontrivial workflow constraints: difficulty labels in the archived validator traces include level values documented as "medium", "hard", and "very_hard" and patterns such as make_before_break_uplink_switchover, paired_link_atomic_mtu_change, and rolling_access_vlan_migration. These labels demonstrate the benchmark included tasks intended to exercise complicated dependency constraints despite producing validator-satisfying outputs under the chosen generation policy.

Missingness, exclusions, and contamination. There were no technical failures affecting the primary analysis: missingness_summary records complete_pairs = 200 and zero failed or unscorable episodes. The contamination detector executed on the preregistered 10% randomized holdout (20 tasks) and found zero disagreements (analysis/contamination_summary.csv). We note a documented reproducibility gap: the explicit randomized selection seed for the Mininet holdout is not labeled in the archived artifacts, which prevents exact deterministic reconstruction of which tasks comprised that holdout despite the contamination summary reporting zero disagreements.

Unexecuted sensitivity branches. Planned sensitivity analyses for alternative contamination treatments and missingness handling were not invoked because no contamination flags or execution failures occurred. Because no repair attempts occurred, we could not empirically evaluate GVR-introduced failure rates or repair-iteration distributions; correspondingly,

the preregistered exploratory analyses returned empty results and are recorded as such in the analysis artifacts.

V. DISCUSSION

Conditional interpretation and statistical mechanics. The confirmatory null (zero baseline validator failures) is an artifact-grounded outcome: baseline_success_rate = 1.0 (200/200) and guarded_success_rate = 1.0 (200/200), producing a paired contingency with $n_{11} = 200$ and no discordant cells. In paired binary inference this configuration is degenerate for McNemar's test—the exact two-sided $p = 1.0$ and the bootstrap percentile 95% CI = [0.0, 0.0] (bootstrap parameters: resamples = 10,000, seed = 1729). Those numbers are not evidence that repair is useless generally; they are a precise statement that, on this curated workload and under the frozen shared-initial generation policy, there were no validator-detectable failures for the GVR pipeline to reduce. The key interpretive point is therefore conditional: the marginal utility of automated repair is a function of baseline validator failure prevalence and validator coverage for the targeted workload.

Artifact-grounded diagnostics explaining the null. Several archived artifacts document why initial candidates passed at scale. First, each task encodes explicit required_sequence and workflow constraints; the primary validator checks those constraints and reports parsed_command_count and required_command_count for each task (representative traces show required_command_count values matching parsed_command_count). Second, difficulty annotations in validator traces (labels such as "make_before_break_uplink_switchover", "paired_link_atomic_mtu_change") show that the task bank exercised nontrivial dependency patterns even when outputs were correct; correctness under these labels indicates the generator/prompt pair produced sequences that satisfied the machine-checkable constraints. Third, adapter accounting (deterministic_reconciliation) shows that hosted_model_call_event_count = 200 and stage_counts = {shared_initial_generation: 200}, confirming a single deterministic generation per task was shared across conditions—this design choice removed sampling variability as a pathway to observe alternative (potentially failing) candidates. Fourth, the preregistered Mininet secondary check ran on a randomized 10% holdout (20 tasks) and the contamination summary (analysis/contamination_summary.csv) records zero disagreements between the primary validator and emulator; this artifact verifies that, in the holdout sample, validator verdicts matched the emulator's checks as executed in this run. We also disclose a reproducibility gap: the labeled artifact set does not include an explicit selection seed for the Mininet holdout, which prevents bit-for-bit reconstruction of the exact holdout selection even though the contamination summary reports zero disagreements.

Methodological tradeoffs: shared-initial pairing versus unconditional failure measurement. The shared_initial paired

design was chosen to isolate the incremental effect of invoking validator+repair on the same generator output. That isolation gains internal control at the cost of observing only the conditional effect on a fixed sample: it cannot capture gains obtainable by re-sampling the LLM (e.g., producing alternative candidates that a repair could act upon) or by employing nonzero temperature sampling to elicit diverse failure modes. For practitioners or evaluators whose goal is to estimate unconditional operational failure rates (the risk of an LLM pipeline producing an unsafe candidate on any application), a design with independent per-condition sampling and multiple draws per task will produce better estimates of unconditional baseline failure prevalence and of the repair mechanism’s ability to convert some failing draws into passes. Our artifact set supports both measurements in principle, but this confirmatory run intentionally prioritized a paired, shared-initial comparison per the preregistered estimand.

Validator coverage and emulator corroboration. Deterministic validators are necessary but not sufficient oracles for operational safety: they codify a set of syntactic and policy checks (here, `deterministic_netops_validator_v5`) that map to task-bank constraints. The Mininet emulator was included as a pragmatic secondary detector to reveal validator blind spots that manifest at the data-plane level; the zero disagreements observed in the 20-task holdout demonstrate no such blind spots were detected in that sample for this run, but holdout scale and the missing labeled selection seed limit how strongly that result can be generalized. A robust operational assurance program should therefore combine diverse validators (vendor-specific semantics, richer device models), larger emulation or testbed sampling, and adversarial/perturbation tests to probe validator blind spots; the artifacts here illustrate where such secondary checks succeeded (no disagreements) and where exact reproducibility of the holdout draw is currently unverifiable (missing seed).

Design implications for guardrail engineering and operational deployment. The neutral outcome suggests practitioners evaluate two complementary axes before investing heavily in automated repair: (1) measure baseline validator failure prevalence on representative workloads under the intended generation policy; (2) assess validator completeness via emulation or selective real-device checks. If baseline failures are already rare under the production generator/prompt policy, then (a) a single-pass repair budget delivers limited marginal safety returns for validator-detected failures, and (b) resources may be better allocated to expanding validator fidelity, adversarial testing, or introducing multi-sample generation and model ensembles to surface latent failure modes. Conversely, when baseline failures are nontrivial, GVR with an appropriately sized repair budget and well-designed repair prompts can reduce validator-detected errors; the archived preregistration captures how repair budgeting and maximum repair calls were constrained here (up to one repair call per task), and the absence of repair events means the exploratory statistics for repair iterations and model-call costs remained empty in the analysis artifacts.

Threats to validity and concrete artifact links. Internal validity is strengthened by preregistration, deterministic adapter enforcement, and deterministic reconciliation artifacts that confirm episode accounting (`complete_pair_count` = 200, `model_calls_used` = 200). Construct validity remains conditional on validator expressiveness: representative `validator_trace` fields (`validator_id`, `validator_version`, `violation_count`, `parsed_command_count`) show what the validator measured, but validators do not represent all device idiosyncrasies. External validity is limited by the single hosted model (`gpt-5-mini`), the `shared_initial` deterministic sampling policy, and the curated task bank; these constraints are recorded in `execution/model_configuration.json` and the preregistration artifact. Conclusion validity: with zero discordant observations the primary confirmatory test is uninformative about the preregistered 50% relative reduction; the observed numbers are nonetheless exact and reproducible within the archived bundle (aside from the Mininet-holdout seed gap noted above).

Practical recommendations for future confirmatory studies. To measure GVR effectiveness where baseline failure prevalence is low, future preregistered runs should (i) adopt a mixed design that includes independent per-condition multi-sample draws to estimate unconditional failure rates; (ii) increase the repair budget and allow multi-step repairs or human-in-the-loop escalation to exercise more repair pathways; (iii) scale secondary emulation checks and record selection seeds for holdouts to enable exact reproducibility; and (iv) evaluate multiple models or ensembles to probe model-dependent brittleness. The archived artifacts and the deterministic reconciliation logs provide a template for implementing these changes while preserving auditability.

VI. LIMITATIONS

- Task-bank scope: the confirmatory sample used a curated synthetic/public intent→configuration bank ($n = 200$); results therefore reflect this bank and may not generalize to broader production workloads or adversarial distributions.
- Single-model and shared-initial setting: only `gpt-5-mini` (hosted) with adapter-enforced shared-initial generation was tested (execution artifacts record `hosted_model_call_event_count` = 200 and `stage_counts` `shared_initial_generation` = 200). Outcomes may differ across model families, independent per-condition sampling, nonzero temperature sampling, or larger repair budgets.
- Validator coverage and oracle limitations: the primary deterministic validator enforces a defined set of syntax/intent constraints; validators can fail to capture real device idiosyncrasies, vendor-specific behaviors, or emergent failure modes detectable only by end-to-end deployment.
- Adversarial robustness untested: this confirmatory run did not include active adversarial injection or red-team tool-description attacks; prior audits indicate such vectors can bypass naive defenses and remain a separate concern.

- Reproducibility gap for contamination check: the randomized holdout selection seed for the Mininet contamination check is absent from the labeled artifacts, preventing exact reconstruction of the holdout selection.
- Single repair budget: the GVR pipeline allowed at most one automated repair prompt per task; multi-step repair strategies, different repair budgets, or human-in-the-loop approvals were not exercised here.

VII. CONCLUSION

We preregistered and executed a sealed paired study comparing LLM-only generation to a deterministic GVR pipeline on 200 NetOps intent→configuration tasks using gpt-5-mini and deterministic_netops_validator_v5. Both pipelines produced validator-passing configurations for every task ($n_{ll} = 200$; no discordant pairs), resulting in an observed paired difference of 0.0, exact McNemar $p = 1.0$, and bootstrap 95% CI = [0.0, 0.0]. The preregistered $\geq 50\%$ relative-reduction hypothesis could not be evaluated because baseline validator failures were absent. The artifact-grounded outcome clarifies that GVR's marginal value is workload dependent: when baseline validator-detectable failures are rare, automated repair yields no measured benefit for those validator-detected errors. Future studies should target workloads with nontrivial baseline failure rates, expand validator fidelity with richer emulation or real-device checks, evaluate multi-sample and multi-model policies, and explore multi-step or human-in-the-loop repair budgets to characterize practical operational gains. All execution artifacts and analysis outputs are archived for independent inspection under the preregistration id cited above.

DISCLOSURE STATEMENT

Disclosure (mandatory). This study executed under the autonomous post-lock preregistration `cnsms2026-gvr-effectiveness-02`. Declared model: gpt-5-mini (provider label: `openai_responses`) invoked via the `hosted_netops_gvr_v1` adapter. The exact initial master prompt is archived at `provenance/master_prompt.txt` (SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acd1f582bb39b15ba). Topic constraints: the human Research Director limited the study to Generative AI and LLMs for NetOps focusing on safety/reliability of generated network changes. Frameworks and tools: orchestration used `hosted_netops_gvr_v1` and the deterministic task generator `deterministic_netops_task_generator_v5`. Primary deterministic validator: `deterministic_netops_validator_v5` (intent/constraint checks). A Mininet-based emulator was preregistered and executed as a secondary disagreement detector on a randomized 10% holdout (20 tasks); the artifact analysis/contamination_summary.csv shows zero disagreements for that holdout. Preregistration and freeze: the experimental plan (estimand, sample size $n = 200$, paired design, stopping rules, max model calls = 400, repair budget = 1 per task, shared-initial generation semantics) was frozen before any model calls. Autonomous execution and artifacts: the run executed autonomously under the

frozen plan (start: 2026-08-30T11:50:34.530059Z; end: 2026-08-30T12:12:07.541650Z); model calls, prompts, seeds, validator traces, emulator traces, logs, and the artifact manifest are archived in the execution bundle. Model-call budget and usage: 200 model calls were used (one shared initial generation per task); up to 200 repair calls were allowed but none were applied because initial candidates passed the validator. Human involvement: the human Research Director selected the topic family and pre-lock constraints; no human manual editing or selective rewriting of technical manuscript content occurred after the autonomy lock. Deviations and restarts: no post-preregistration design changes or technical restarts occurred. Known reproducibility gap: the explicit randomized selection seed used to draw the Mininet 10% holdout is not present as a labeled artifact in the archive; this absence prevents exact deterministic reconstruction of the drawn holdout despite the contamination summary reporting no disagreements. The master prompt archive reference above is the immutable prompt required by the track.

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